

Multivariate Optimisation - Solutions

LLE – Mathematics and Statistics Skills

1. Finding maximum profit and revenue

(a) $\Pi = TR - TC = 80Q - 100 - Q^2$

(b) $\frac{d\Pi}{dQ} = 0$ or $MR = MC$

$$\implies 80 = 2Q$$

$$\implies Q = 40$$

(c) $\Pi = 80(40) - 100 - (40)^2 = 1500$

(d)

$$\frac{d\Pi}{dQ} = 80 - 2Q$$

$$\frac{d^2\Pi}{dQ^2} = -2 < 0 \implies \text{maximum}$$

2. $z(x, y) = x^2 + y^2$

(a) $\frac{\partial z}{\partial x} = 2x, \frac{\partial z}{\partial y} = 2y$

(b)

$$\frac{\partial z}{\partial x} = 0$$

$$\implies 2x = 0$$

$$\implies x = 0$$

$$\frac{\partial z}{\partial y} = 0$$

$$\implies 2y = 0$$

$$\implies y = 0$$

$$z = 0^2 + 0^2 = 0 \implies (0, 0, 0)$$

(c)

$$\frac{\partial^2 z}{\partial x^2} = 2$$

$$\frac{\partial^2 z}{\partial y^2} = 2$$

$$\frac{\partial^2 z}{\partial x \partial y} = 0$$

$$\frac{\partial^2 z}{\partial x^2} \frac{\partial^2 z}{\partial x^2} - \left(\frac{\partial^2 z}{\partial x \partial y} \right)^2 = 4$$

Since $\frac{\partial^2 z}{\partial x^2} \frac{\partial^2 z}{\partial x^2} - \left(\frac{\partial^2 z}{\partial x \partial y} \right)^2 = 4 > 0$ and $\frac{\partial^2 z}{\partial x^2} = 2 > 0$ then $(0, 0, 0)$ is a local minimum

3. (a) $z = x^2 - y^2 + 20$

$$\frac{\partial z}{\partial x} = 2x$$

$$\frac{\partial z}{\partial y} = -2y$$

$$2x = 0 \implies x = 0$$

$$2y = 0 \implies y = 0$$

Coordinates $(0, 0, 20)$

$$\frac{\partial^2 z}{\partial x^2} = 2 \quad \frac{\partial^2 z}{\partial y^2} = -2 \quad \frac{\partial^2 z}{\partial x \partial y} = 0$$

$$\frac{\partial^2 z}{\partial x^2} \frac{\partial^2 z}{\partial x^2} - \left(\frac{\partial^2 z}{\partial x \partial y} \right)^2 = -4 < 0$$

So $(0, 0, 20)$ is a saddle point

(b) $z = 20x - x^2 + 6y - y^2$

$$\frac{\partial z}{\partial x} = 20 - 2x \quad \frac{\partial z}{\partial y} = 6 - 2y$$

$$20 - 2x = 0 \implies x = 10$$

$$6 - 2y = 0 \implies y = 3$$

Coordinates (10, 3, 109)

$$\begin{aligned} \frac{\partial^2 z}{\partial x^2} &= -2 < 0 & \frac{\partial^2 z}{\partial y^2} &= -2 & \frac{\partial^2 z}{\partial x \partial y} &= 0 \\ \frac{\partial^2 z}{\partial x^2} \frac{\partial^2 z}{\partial x^2} - \left(\frac{\partial^2 z}{\partial x \partial y} \right)^2 &= 4 > 0 \end{aligned}$$

So (10, 3, 109) is a local maximum

(c) $z = 2x^2 + 4x - 2xy + y^2$

$$\frac{\partial z}{\partial x} = 4x + 4 - 2y = 0 \quad \frac{\partial z}{\partial y} = -2x + 2y = 0$$

$$4x - 2y = -4 \tag{1}$$

$$-2x + 2y = 0 \tag{2}$$

$$2x = -4 \quad (1) + (2)$$

$$x = -2, \text{ equation (2)} \implies y = x, \text{ therefore } y = -2$$

Coordinates (-2, -2, -4)

$$\begin{aligned} \frac{\partial^2 z}{\partial x^2} &= 4 > 0 & \frac{\partial^2 z}{\partial y^2} &= 2 & \frac{\partial^2 z}{\partial x \partial y} &= -2 \\ \frac{\partial^2 z}{\partial x^2} \frac{\partial^2 z}{\partial x^2} - \left(\frac{\partial^2 z}{\partial x \partial y} \right)^2 &= 4 > 0 \end{aligned}$$

So (-2, -2, -4) is a local minimum

4. $z = x^2y - 2xy^2 + 3xy - 1$

$$\frac{\partial z}{\partial x} = 2xy - 2y^2 + 3y = y(2x - 2y + 3) = 0$$

$$\frac{\partial z}{\partial y} = x^2 - 4xy + 3x = x(x - 4y + 3) = 0$$

From the $\frac{\partial z}{\partial x} = 0$, we could have $y = 0$

Substitute this into $\frac{\partial z}{\partial y} = 0$, then

$$x(x + 3) = 0 \implies x = 0, x = -3$$

$$(0, 0, -1), (-3, 0, -1)$$

From the $\frac{\partial z}{\partial y} = 0$, we could have $x = 0$

Substitute this into $\frac{\partial z}{\partial x} = 0$, then

$$y(-2y + 3) = 0 \implies y = 0, y = \frac{3}{2}$$

$$(0, 0, -1), \left(0, \frac{3}{2}, -1\right)$$

We could also have both:

$$2x - 2y + 3 = 0$$

$$x - 4y + 3 = 0$$

The top equation minus twice the bottom equation gives:

$$6y - 3 = 0 \implies y = \frac{1}{2}, x = -1, z = -\frac{3}{2}$$

All coordinates:

$$(0, 0, -1), (-3, 0, -1), \left(0, \frac{3}{2}, -1\right), \left(-1, \frac{1}{2}, -\frac{3}{2}\right)$$

$$5. AC = Q_1^2 + 2Q_2^2 - 8Q_1 - 12Q_2 + 100$$

$$\frac{\partial AC}{\partial Q_1} = 2Q_1 - 8 = 0$$

$$\frac{\partial AC}{\partial Q_2} = 4Q_2 - 12 = 0$$

From these two equations we get $Q_1 = 4$, $Q_2 = 3$, $AC = 66$

$$\frac{\partial^2 AC}{\partial Q_1^2} = 2 > 0$$

$$\frac{\partial^2 AC}{\partial Q_2^2} = 4$$

$$\frac{\partial^2 AC}{\partial Q_1 \partial Q_2} = 0$$

$$D = \left(\frac{\partial^2 AC}{\partial Q_1^2} \right) \left(\frac{\partial^2 AC}{\partial Q_2^2} \right) - \left(\frac{\partial^2 AC}{\partial Q_1 \partial Q_2} \right)^2 = 8 > 0$$

$$D > 0 \text{ and } \frac{\partial^2 AC}{\partial Q_1^2} > 0 \implies \text{minimum}$$